

Research Notes on Dropout regularization

Dropout regularization

$$\tilde{h} = r \odot h, \quad r_i \sim \text{Bernoulli}(p)$$

>> Section 1

This can be viewed as inducing a regularizer over the L_2 norm over members of each group followed by an L_1 norm over groups. This can be solved by the proximal method, where the proximal operator is a block-wise soft-thresholding function:

>> Section 2

Enforcing a sparsity constraint on w can lead to simpler and more interpretable models. This is useful in many real-life applications such as computational biology. An example is developing a simple predictive test for a disease in order to minimize the cost of performing medical tests while maximizing predictive power. A sensible sparsity constraint is the L_0 norm $\|w\|_0$, defined as the number of non-zero elements in w . Solving a L_0 regularized learning problem, however, has been demonstrated to be NP-hard. The L_1 norm (see also Norms) can be used to approximate the optimal L_0 norm via convex relaxation. It can be shown that the L_1 norm induces sparsity. In the case of least squares, this problem is known as LASSO in statistics and basis pursuit in signal processing.

>> Section 3

where the third statement is a first-order condition. By construction of the optimization problem, other values of w give larger values for the loss function. This can be verified by examining the second derivative $\nabla^2 w$. During training, this algorithm takes $O(d^3 + nd^2)$ time. The terms correspond to the matrix inversion and calculating $X^T X$, respectively. Testing takes $O(nd)$ time.

>> Section 4

$R(w) = \sum_{g=1}^G \|w_g\|_2^2$, where $\|w_g\|_2^2 = \sum_{j=1}^{|G_g|} (w_{gj})^2$

>> Section 5

Regularization in machine learning In machine learning, a key challenge is enabling models to accurately predict outcomes on unseen data, not just on familiar training data. Regularization is crucial for addressing overfitting—where a model memorizes training data details but cannot generalize to new data. The goal of regularization is to encourage models to learn the broader patterns within the data rather than memorizing it. Techniques like early stopping, L1 and L2 regularization, and dropout are designed to prevent overfitting and underfitting, thereby enhancing the model's ability to adapt to and perform well with new data, thus improving model generalization.

>> Section 6

Group sparsity without overlaps Groups of features can be regularized by a sparsity constraint, which can be useful for expressing certain prior knowledge into an optimization problem. In the case of a linear model with non-overlapping known groups, a regularizer can be defined:

>> Section 7

In mathematics, statistics, finance, and computer science, particularly in machine learning and inverse problems, regularization is a process that converts the answer to a problem to a simpler one. It is often used in solving ill-posed problems or to prevent overfitting. Although regularization procedures can be divided in many ways, the following delineation is particularly helpful:

>> Section 8

This regularizer constrains the functions learned for each task to be similar to the overall average of the functions across all tasks. This is useful for expressing prior information that each task is expected to share with each other task. An example is predicting blood iron levels measured at different times of the day, where each task represents an individual.

>> Section 9

$\min_{w \in \mathbb{R}^{p \times 1 \times n}} \|Xw - Y\|_2^2 + \lambda \|w\|_1$

>> Section 10

$w_T = \gamma \sum_{i=0}^{T-1} (I - \gamma X^T X)^i X^T Y$

>> Section 11

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$R(f) = \bar{f}^T L \bar{f}$ where $L = D - W$ is the Laplacian matrix of the graph induced by W . The optimization problem $\min_{f \in \mathbb{R}^m} R(f)$, $m = u + l$ can be solved analytically if the constraint $f(x_i) = y_i$ is applied for all supervised samples. The labeled part of the vector f is therefore obvious. The unlabeled part of f is solved for by:

>> Section 12

$\min_{w \in \mathbb{R}^p} \frac{1}{n} \|Xw - Y\|^2 + \lambda (\alpha \|w\|_1 + (1 - \alpha) \|w\|_2^2)$, $\alpha \in [0, 1]$

>> Section 13

Regularizers for multitask learning In the case of multitask learning, T problems are considered simultaneously, each related in some way. The goal is to learn T functions, ideally borrowing strength from the relatedness of tasks, that have predictive power. This is equivalent to learning the matrix $W : T \times D$.

>> Section 14

$I_S[f_n] = \frac{1}{n} \sum_{i=1}^n V(f_n(x^i), y^i)$

>> Section 15

$w_0 = 0$ $w_{t+1} = (I - \gamma_n X^T X) w_t + \gamma_n X^T Y$

$$\begin{aligned} w_{t+1} &= \left(I - \frac{\gamma_n}{\|X\|^2} X^T X \right) w_t + \frac{\gamma_n}{\|X\|^2} X^T Y \end{aligned}$$

>> Section 16

$S_\lambda(v) = \begin{cases} v - \lambda, & \text{if } v > \lambda \\ 0, & \text{if } v \in [-\lambda, \lambda] \\ v + \lambda, & \text{if } v < -\lambda \end{cases}$